

DATA FLOW DIAGRAM (DFD)

Project Name:

ComicCraft – AI Powered Comic Story Generator

1. Introduction

The Data Flow Diagram (DFD) represents how information moves through the ComicCraft system. It illustrates the interaction between users, AI services, backend processes, data storage, and output generation modules.

ComicCraft converts user-provided story prompts into complete AI-generated comics by integrating story generation, image generation, comic layout creation, and PDF export functionalities.

2. External Entities

E1: User

The user interacts with the ComicCraft platform by:

- Entering story prompts
- Selecting character details
- Choosing tone and art style
- Viewing generated comics
- Downloading PDF comics

E2: Gemini AI Services

External AI service responsible for:

- Comic outline generation
- Story generation
- Dialogue generation
- Narration generation

E3: Image Generation Service

External AI image generation model responsible for:

- Comic panel illustration generation
- Character visualization
- Scene creation

3. Processes

Process 1: Input Collection & Validation

Collects user input and validates:

- Story prompt
- Character name
- Tone
- Setting
- Art style

Process 2: Story Outline Generation

Uses Gemini Flash to generate:

- 5-panel comic structure
- Scene descriptions
- Story progression

Process 3: Detailed Story Generation

Uses Gemini Pro to generate:

- Narration
- Dialogues
- Captions
- Detailed panel content

Process 4: Image Generation

Generates illustrations for each panel using AI image-generation models.

Process 5: Comic Layout Builder

Combines:

- Generated images
- Narration
- Dialogues
- Captions

into a comic format.

Process 6: PDF Generation

Converts the comic layout into a downloadable PDF document.

4. Data Stores

D1: User Input Data

Stores:

- Story prompts
- Character details
- Tone
- Art style preferences

D2: Generated Story Data

Stores:

- Story outline
- Dialogues
- Narration
- Captions

D3: Generated Images

Stores:

- Comic panel illustrations
- Character images
- Scene images

D4: Comic PDF Repository

Stores:

- Generated comic PDF files
-

5. Data Flow Description

Flow 1

User → Input Collection & Validation

Data:

- Story Prompt
- Character Name
- Tone

- Setting
- Art Style

Flow 2

Input Collection & Validation → Story Outline Generation

Data:

- Validated User Input

Flow 3

Story Outline Generation ↔ Gemini AI Service

Data:

- Story Prompt
- Comic Outline

Flow 4

Story Outline Generation → Detailed Story Generation

Data:

- Panel Outline

Flow 5

Detailed Story Generation ↔ Gemini AI Service

Data:

- Narration
- Dialogue
- Captions

Flow 6

Detailed Story Generation → Generated Story Data

Data:

- Complete Comic Script

Flow 7

Generated Story Data → Image Generation

Data:

- Scene Descriptions
- Image Prompts

Flow 8

Image Generation ↔ Image Generation Service

Data:

- Image Prompts
- Generated Illustrations

Flow 9

Image Generation → Generated Images

Data:

- Comic Panel Images

Flow 10

Generated Story Data + Generated Images → Comic Layout Builder

Data:

- Dialogues
- Narration
- Captions
- Illustrations

Flow 11

Comic Layout Builder → PDF Generation

Data:

- Final Comic Layout

Flow 12

PDF Generation → Comic PDF Repository

Data:

- PDF Comic

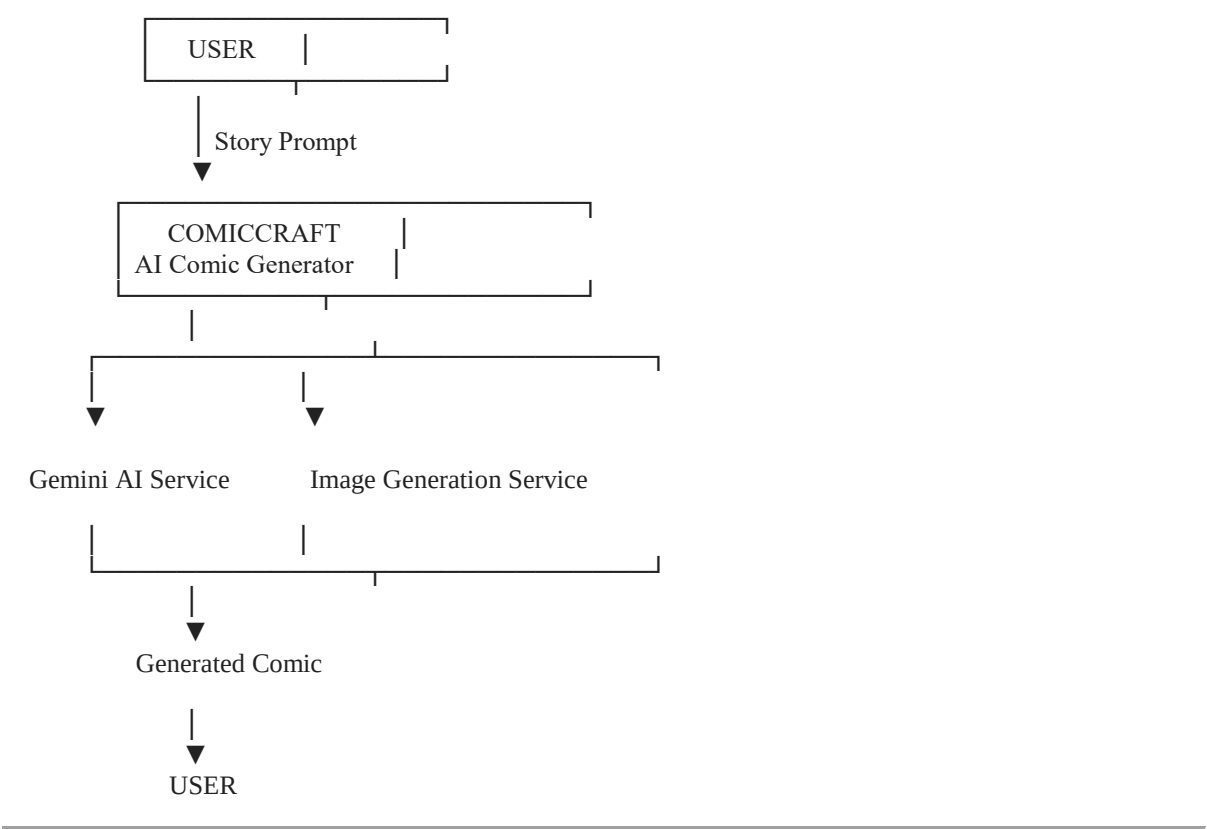
Flow 13

Comic PDF Repository → User

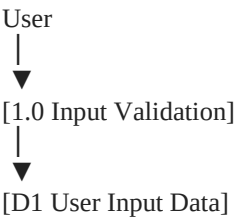
Data:

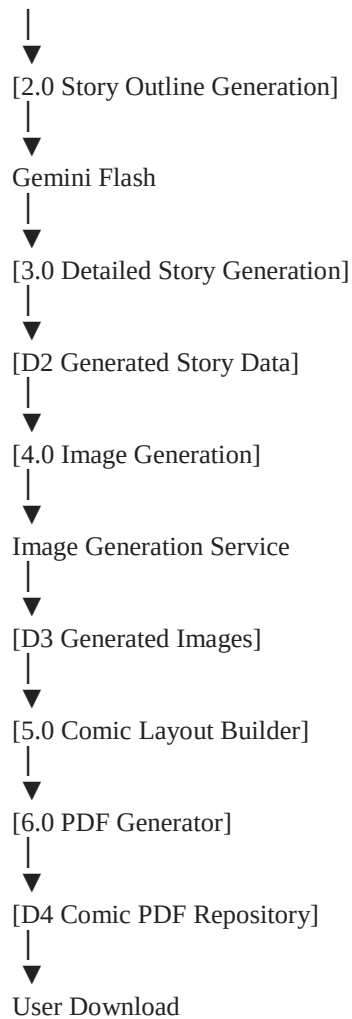
- Downloadable Comic PDF

6. DFD Level-0 (Context Diagram)



7. DFD Level-1





8. Conclusion

The Data Flow Diagram illustrates the movement of data within ComicCraft from user input to final PDF comic generation. The system integrates multiple AI services to automate storytelling, illustration creation, comic assembly, and PDF export. The DFD provides a clear understanding of how information is processed, stored, and delivered to users throughout the comic generation workflow.